

Research Plan Workshop

PS50008A Week 13 | Lab Walkthrough

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Welcome to Lab

Research Plan Workshop

Today's Focus

Building on Wednesday's lecture. Today you will:

- Operationalise your IV and DV for your chosen topic
- Write and justify hypotheses (one-tailed vs two-tailed)
- Decide on experimental design (between vs within)
- Match design to appropriate statistical test
- Begin drafting your Method section

Lab Structure (~120 min)

Time	Activity
10:00-10:10	These slides + overview
10:10-10:40	Activity 1: Operationalisation Workshop
10:40-11:05	Activity 2: Hypothesis Writing
11:05-11:15	BREAK
11:15-11:40	Activity 3: Design & Analysis Matching
11:40-12:00	Activity 4: Method Section Draft

Key Differences from Previous Labs

Workshop Format

Today is a structured **writing workshop**. Time for you to make progress with your research plans.

Instructors are available to help, and happy to do so.

Remember, to get the advantage of their advice, you have to have a go!

We want you to start confidently and in plenty of time!

Activity 1: Operationalisation

30 minutes

The Three Topics

The Research Plan topics:

1. **Font difficulty → Memory**

- Does harder-to-read text improve retention?

2. **Hydration → Cognitive performance**

- Does drinking water improve thinking?

3. **Eye contact → Perceived trustworthiness**

- Does eye contact affect trust perceptions?

Key Question for Your IV

How EXACTLY will your conditions differ?

Font study: What fonts? What sizes? What colours?

Hydration study: How much water? What's the control?

Eye contact study: Photos or videos? What angle of gaze?

Common Errors to Avoid

Watch for these issues in IV operationalisation:

- **Too vague:** “One group saw hard font, one saw easy font”
- **Multiple differences:** Font AND size AND colour all change
- **No control details:** Only describing experimental condition
- **Unmeasurable:** “Participants felt more stressed”

Key Question for Your DV

How EXACTLY will you measure the outcome?

You need to specify:

- The task/measure used
- The scoring method
- The possible range of scores

The Replicability Test

Ask yourself:

“Could a stranger run my study EXACTLY from my description?”

If the answer is “maybe” or “no” — more detail is needed.

Activity 1: Work Time

Complete questions 1-5 on the Activity page.

Write out your IV and DV operationalisations in full.

19:59

Activity 2: Hypothesis Writing

25 minutes

Key Teaching Points

One-tailed vs Two-tailed: Not about preference, but about evidence

You must understand:

- One-tailed = strong prior evidence for direction
- Two-tailed = uncertainty about direction
- Choice must be JUSTIFIED with citations

Questions to Ask Yourself

1. “What does prior research say about my topic?”
2. “Is there strong evidence for a specific direction?”
3. “What would I conclude if the effect went the OTHER way?”

Common Misconceptions

- “One-tailed is better because it’s more powerful”
- “Two-tailed is always safer”
- “I can decide after seeing my data”
- “It doesn’t really matter”

All of these are problematic! Do you know why?

Example Justifications

One-tailed (appropriate):

“Diemand-Yauman et al. (2011) found disfluent fonts improved recall, and this has been replicated. I predict disfluent > fluent.”

Two-tailed (appropriate):

“Replication attempts have been mixed. I predict a difference but won’t specify direction.”

Activity 2: Work Time

Complete the hypothesis section on the Activity page.

Write both H_0 and H_1 , then justify your choice of direction.

14:59

BREAK

09:59

Activity 3: Design Matching

25 minutes

The Critical Link

Design → Data Structure → Statistical Test

This chain of logic is REQUIRED in your Research Plan.

You must explain WHY your design leads to your test choice.

What Would Each Design Look Like?

For each topic, articulate:

Topic	Between-Subjects	Within-Subjects
Font	Group A: fluent; Group B: disfluent	Same people read BOTH
Hydration	Group A: water; Group B: no water	Same person tested twice
Eye contact	Different photos for different people	Same person rates BOTH gaze types

Design Trade-offs

Design	Pro	Con
Between	No order effects	Individual differences
Within	Controls individual differences	Order effects

The Test Selection Rule

Between-subjects → Independent samples t-test

- Two separate groups
- Scores are unrelated

Within-subjects → Paired samples t-test

- Same people twice
- Scores are paired

Common Error

Watch for this mistake

Stating the test WITHOUT justifying why.

Wrong: “I will use an independent t-test.”

Right: “Because different participants are in each condition, the data will consist of two independent groups. An independent samples t-test is therefore appropriate.”

Activity 3: Work Time

Complete Activity 3 on the Activity page.

Write out BOTH designs for your topic, identify trade-offs, then make your choice with justification.

14:59

Activity 4: Method Draft

20 minutes

Method Section Structure

You should draft these subsections:

1. **Design:** Type, IV with levels, DV with measurement
2. **Participants:** N, recruitment, assignment
3. **Materials:** Stimuli, measures, equipment
4. **Procedure:** Step-by-step what happens

What to Check in Your Draft

As you write, make sure you have:

- Sufficient operationalisation detail
- Clear design-to-test justification
- Replicable procedure description
- Appropriate sample size (typically $n = 30$ per group)

Activity 4: Work Time

Draft your Method section using the templates on the Activity page.

14:59

Peer Feedback (if time permits)

Swap drafts with a partner and answer:

1. Could you replicate this study?
2. What details are missing?
3. Is the design-test link clear?

Wrap-Up

Key Messages

! Before you leave:

1. **Reading Week** = Time to write your Research Plan draft
2. **Operationalisation** = The foundation of a good plan
3. **Justification** = Required for hypotheses AND test choice
4. **Due date** = Friday 27 February, Week 16

Next Week

Week 14: Is Your Effect Real?

- NHST logic and p-values
- Confidence intervals and effect sizes
- t-tests in practice
- Complete inferential toolkit

Questions?